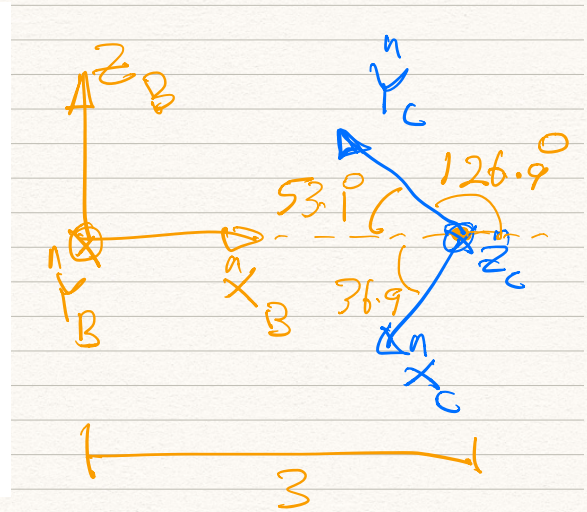
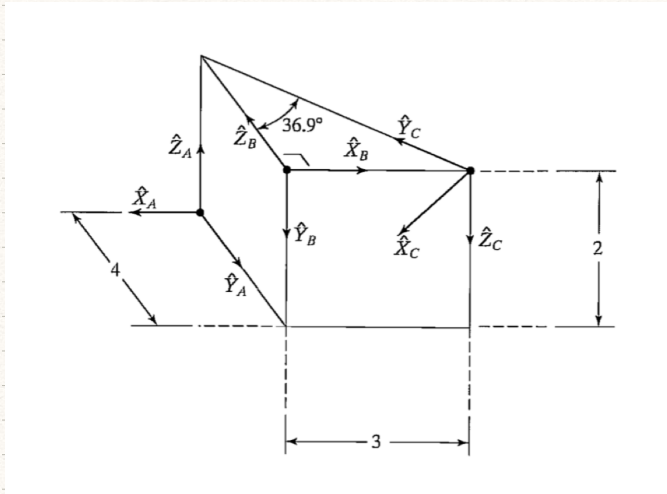
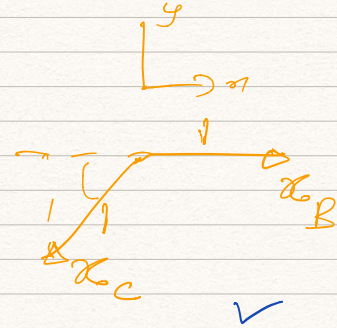
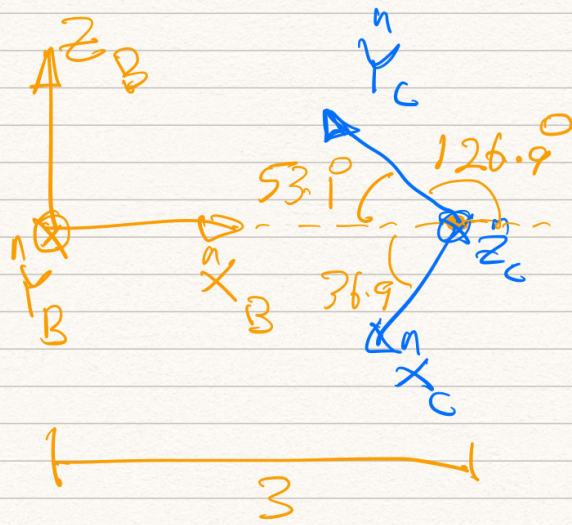




$${}^B R_C = \begin{matrix} & x_C & y_C & z_C \\ \begin{matrix} x_B \\ y_B \\ z_B \end{matrix} & \begin{bmatrix} x_C \cdot x_B & y_C \cdot x_B & z_C \cdot x_B \\ x_C \cdot y_B & y_C \cdot y_B & z_C \cdot y_B \\ x_C \cdot z_B & y_C \cdot z_B & z_C \cdot z_B \end{bmatrix} \end{matrix}$$

$${}^B P_C = \begin{bmatrix} 3 \\ 0 \\ 0 \end{bmatrix} \quad \text{--- (2)}$$

$${}^B T_C = \begin{bmatrix} {}^B R_C & {}^B P_C \\ 0_{3 \times 3} & 1 \end{bmatrix} \quad \text{--- (3)}$$



$${}^B R_C = \begin{bmatrix} -0.7997 & -0.6004 & 0 \\ 0 & 0 & 1 \\ -0.6004 & 0.7997 & 0 \end{bmatrix}$$

Example:

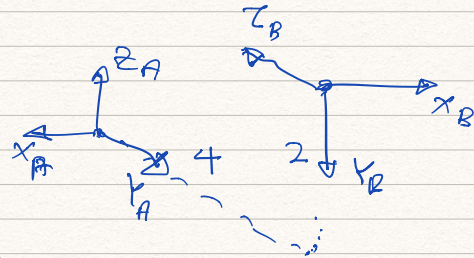
$$\begin{aligned} x_C \cdot x_B &= |x_C| |x_B| \\ \text{or:} \quad &\cos(180 + 36.9) \\ &= 1 \times 1 \times (-0.7997) \\ &= -0.7997 \end{aligned}$$

Therefore

$${}^B T_C = \begin{matrix} & \begin{matrix} x_C & y_C & z_C \end{matrix} \\ \begin{matrix} x_B \\ y_B \\ z_B \end{matrix} & \begin{bmatrix} -0.8 & -0.6 & 0 \\ 0 & 0 & 1 \\ -0.6 & 0.8 & 0 \end{bmatrix} \end{matrix} \begin{matrix} \begin{matrix} 3 \\ 0 \\ 0 \end{matrix} \\ \end{matrix}$$

$${}^A R_B = \begin{bmatrix} x_B \cdot x_A & y_B \cdot x_A & z_B \cdot x_A \\ x_B \cdot y_A & y_B \cdot y_A & z_B \cdot y_A \\ x_B \cdot z_A & y_B \cdot z_A & z_B \cdot z_A \end{bmatrix}$$

$${}^A P_B = \begin{bmatrix} 0 \\ 4 \\ 2 \end{bmatrix}$$



$${}^A R_B = \begin{bmatrix} x_B & y_B & z_B \\ x_A & -1 & 0 & 0 \\ y_A & 0 & 0 & -1 \\ z_A & 0 & -1 & 0 \end{bmatrix}$$

$${}^A T_B = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 4 \\ 2 \\ 1 \end{bmatrix}$$

Therefore

$$K^T C = A^T B \times B^T C$$

$$= \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 4 \\ 0 & -1 & 0 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -0.8 & -0.6 & 0 & 3 \\ 0 & 0 & 1 & 0 \\ -0.6 & 0.8 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{matrix} A_{RC} & A_{PC} \end{matrix} \begin{bmatrix} 0.8 & 0.6 & 0 \\ 0.6 & -0.8 & 0 \\ 0 & 0 & -1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} -3 \\ 4 \\ 2 \\ 1 \end{bmatrix}$$

Question-2

- Choice - 1 is correct

Question - 3

- d is correct

Question - 4

- a - correct

Question-5

a - is correct

Question - 6

b - is correct

Question - 7

d - is correct

~~Question - 8~~

~~b - is correct~~

Question - 8

b - is correct

Question - 10

$\begin{pmatrix} x \\ y \\ z \end{pmatrix}$

Fixed frame

x - y - z ✓

y - z - x ✓

z - x - y ✓

x - z - y

z - y - x ✓

y - x - z

In addition, moving frames provide 6-mov.

Z-X-Z

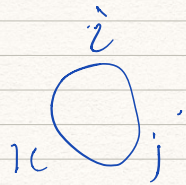
Y-X-Y

Z-Y-Z

X-Z-X

X-Y-X

X-Z-X



$$(c_{\frac{\theta}{2}} + k s_{\frac{\theta}{2}})(i x + j y)(c_{\frac{\theta}{2}} - k s_{\frac{\theta}{2}})$$

$$= [i x c_{\frac{\theta}{2}} + j y c_{\frac{\theta}{2}} + j x s_{\frac{\theta}{2}} - i y s_{\frac{\theta}{2}}](c_{\frac{\theta}{2}} - k s_{\frac{\theta}{2}})$$

$$= [i(x c_{\frac{\theta}{2}} - y s_{\frac{\theta}{2}}) + j(x s_{\frac{\theta}{2}} + y c_{\frac{\theta}{2}})](c_{\frac{\theta}{2}} - k s_{\frac{\theta}{2}})$$

$$= i(x c_{\frac{\theta}{2}}^2 - y s_{\frac{\theta}{2}} c_{\frac{\theta}{2}}) + j(x s_{\frac{\theta}{2}} c_{\frac{\theta}{2}} + y c_{\frac{\theta}{2}}^2) + j(x c_{\frac{\theta}{2}} s_{\frac{\theta}{2}} - y s_{\frac{\theta}{2}}^2) + i(x s_{\frac{\theta}{2}}^2 + y c_{\frac{\theta}{2}} s_{\frac{\theta}{2}})$$

$$= i(x(c_{\frac{\theta}{2}}^2 - s_{\frac{\theta}{2}}^2) - 2y s_{\frac{\theta}{2}} c_{\frac{\theta}{2}})$$

$$+ j(2x s_{\frac{\theta}{2}} c_{\frac{\theta}{2}} + y(c_{\frac{\theta}{2}}^2 - s_{\frac{\theta}{2}}^2))$$

$$= i \left(x \cos \theta - y \sin \theta \right) + j \left(x \sin \theta + y \cos \theta \right)$$

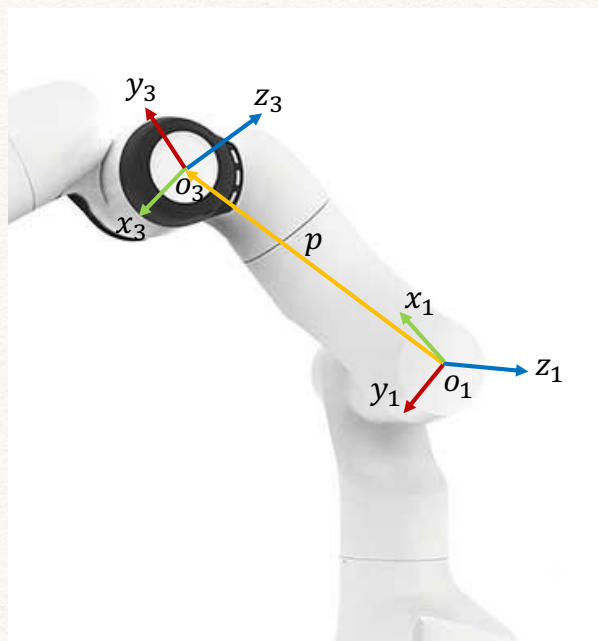
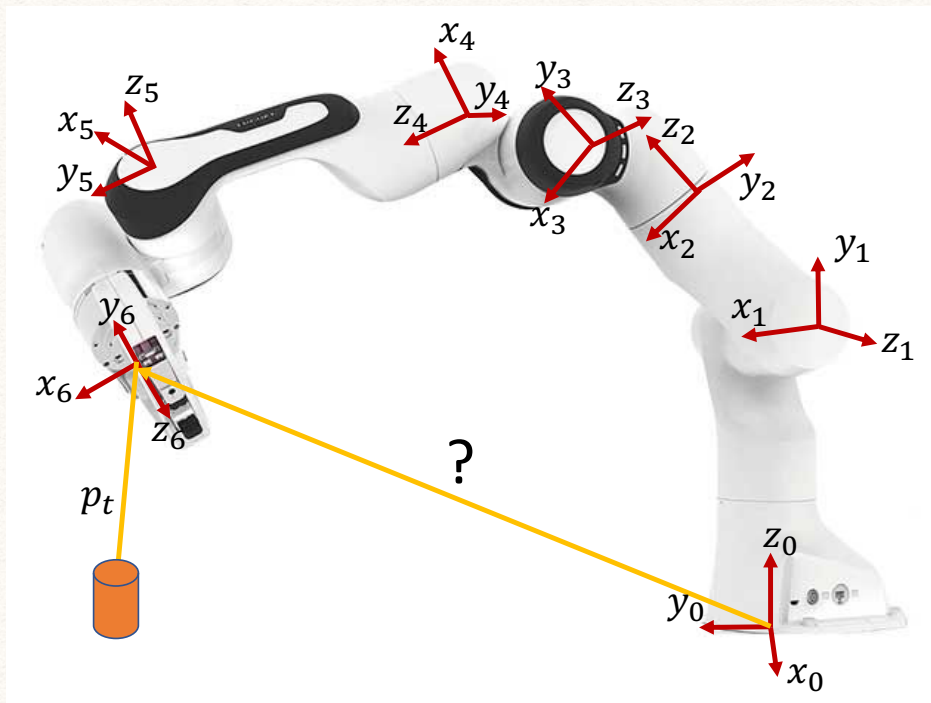
$$= \underbrace{\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}}_{R} \begin{pmatrix} x \\ y \end{pmatrix}$$

3D Kinematics tutorial-1

Learning outcomes:

- Know what the forward kinematics problem is
- Know how to assign frames to joints in a 3D robot.
- Know what a homogeneous coordinate system is.
- Know how to interpret a homogeneous transformation matrix.

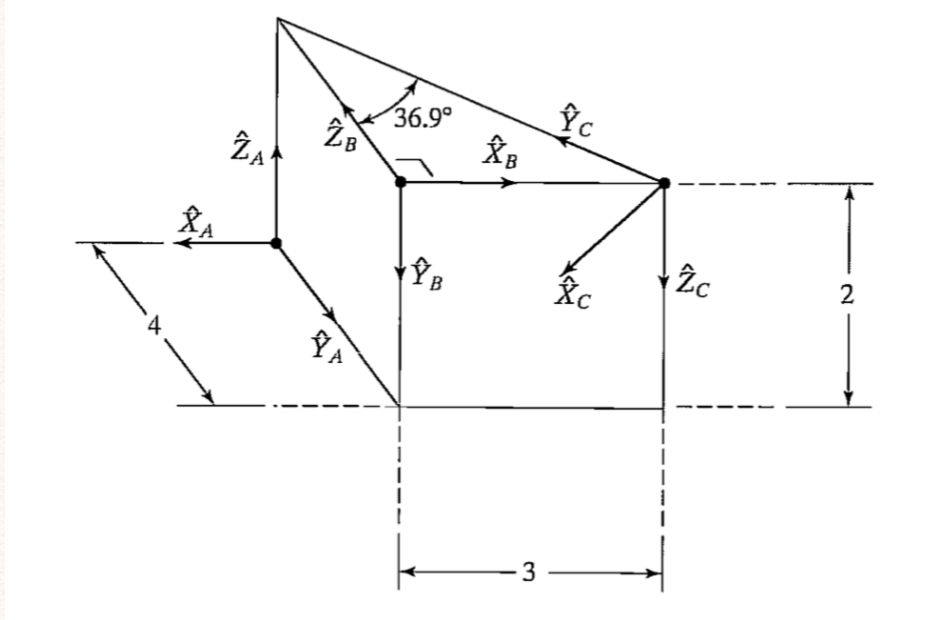
The forward kinematics problem Given knowledge of only the joint angles and link lengths, how to determine the position and orientation of the end-effector of a robot relative the the base.



A coordinate frame i consists of an origin, denoted O_i , and three mutually orthogonal basis vectors, denoted (x_i, y_i, z_i) , that are fixed within a given body.

Question-1

Craig, Q 2.32 Find the transformation matrices ${}^B T_C$, ${}^A T_B$, and ${}^A T_C$.



Question-2

You found a piece of paper with a set of rotation matrices. However, some of the numbers are not clear. Which of the following rotation matrices could be the complete set?

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ ? & ? & ? \end{bmatrix} \quad \begin{bmatrix} ? & 0 & 0.7071 \\ ? & 1 & 0 \\ ? & 0 & 0.7071 \end{bmatrix} \quad \begin{bmatrix} 0.866 & ? & 0 \\ 0.5 & ? & 0 \\ 0 & ? & 1 \end{bmatrix}$$

a)

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} \quad \begin{bmatrix} 0.7071 & 0 & 0.7071 \\ 0 & 1 & 0 \\ -0.7071 & 0 & 0.7071 \end{bmatrix} \quad \begin{bmatrix} 0.866 & -0.5 & 0 \\ 0.5 & 0.866 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

b)

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & 0 \end{bmatrix} \quad \begin{bmatrix} 0.7071 & 0 & 0.7071 \\ 0 & 1 & 0 \\ 0 & 0 & 0.7071 \end{bmatrix} \quad \begin{bmatrix} 0.866 & 0 & 0 \\ 0.5 & 0.866 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

c)

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & 0 \end{bmatrix} \quad \begin{bmatrix} -0.7071 & 0 & 0.7071 \\ 0 & 1 & 0 \\ -0.7071 & 0 & 0.7071 \end{bmatrix} \quad \begin{bmatrix} 0.866 & 0.5 & 0 \\ 0.5 & 0.866 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

d)

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} \quad \begin{bmatrix} -0.7071 & 0 & 0.7071 \\ 0 & 1 & 0 \\ 1 & 0 & 0.7071 \end{bmatrix} \quad \begin{bmatrix} 0.866 & -0.5 & 0 \\ 0.5 & -0.866 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Question-3

Which of the following give the correct expressions for rotation angles given the following rotation matrix?

$${}^i\mathbf{R}_j(\alpha, \beta, \gamma) = \begin{bmatrix} cac\beta c\gamma - sas\gamma & -cac\beta s\gamma - sac\gamma & cas\beta \\ sac\beta c\gamma - cas\gamma & -sac\beta s\gamma + cac\gamma & sas\beta \\ -s\beta c\gamma & s\beta s\gamma & c\beta \end{bmatrix} \quad (1)$$

$$= \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}$$

a)

$$\beta = \text{atan2}(\sqrt{r_{33}^2 + r_{32}^2}, r_{31}); \alpha = \text{atan2}\left(\frac{r_{32}}{s\beta}, -\frac{r_{31}}{s\beta}\right); \gamma = \text{atan2}\left(\frac{r_{23}}{s\beta}, \frac{r_{13}}{s\beta}\right) \quad (2)$$

b)

$$\beta = \text{atan2}(r_{31}, \sqrt{r_{32}^2 + r_{33}^2}); \alpha = \text{atan2}\left(\frac{r_{21}}{c\beta}, \frac{r_{11}}{c\beta}\right); \gamma = \text{atan2}\left(\frac{r_{32}}{c\beta}, \frac{r_{33}}{c\beta}\right) \quad (3)$$

c)

$$\beta = \text{atan2}(r_{31}, \sqrt{r_{32}^2 + r_{33}^2}); \alpha = \text{atan2}\left(\frac{r_{23}}{s\beta}, \frac{r_{13}}{s\beta}\right); \gamma = \text{atan2}\left(\frac{r_{32}}{s\beta}, -\frac{r_{31}}{s\beta}\right) \quad (4)$$

d)

$$\beta = \text{atan2}(\sqrt{r_{31}^2 + r_{32}^2}, r_{33}); \alpha = \text{atan2}\left(\frac{r_{23}}{s\beta}, \frac{r_{13}}{s\beta}\right); \gamma = \text{atan2}\left(\frac{r_{32}}{s\beta}, -\frac{r_{31}}{s\beta}\right) \quad (5)$$

Question-4

"A rotation of a frame in 3D space around an axis passing through its origin can be realized through 3 successive rotations around any 3 linearly independent axes". Is this statement correct?

a) Correct

b) Not correct

Question-5

The homogeneous transformation matrix from the end point frame to the base frame of a 3-link manipulator is given by

$${}^0\mathbf{T}_4 = \begin{bmatrix} \cos(\theta_1 + \theta_2 + \theta_3) & -\sin(\theta_1 + \theta_2 + \theta_3) & 0 & L_1 \cos(\theta_1) + L_2 \cos(\theta_1 + \theta_2) \\ \sin(\theta_1 + \theta_2 + \theta_3) & \cos(\theta_1 + \theta_2 + \theta_3) & 0 & L_1 \sin(\theta_1) + L_2 \sin(\theta_1 + \theta_2) \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (6)$$

The Jacobian matrix of the above manipulator is given by

a)

$$\mathbf{J} = \begin{bmatrix} -L_2 \sin(\theta_1 + \theta_2) - L_1 \sin(\theta_1) & -L_2 \sin(\theta_1 + \theta_2) & 0 \\ L_2 \cos(\theta_1 + \theta_2) + L_1 \cos(\theta_1) & L_2 \cos(\theta_1 + \theta_2) & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad (7)$$

b)

$$\mathbf{J} = \begin{bmatrix} -L_2 \sin(\theta_1 + \theta_2) - L_1 \sin(\theta_1) & 0 & -L_2 \sin(\theta_1 + \theta_2) \\ L_2 \cos(\theta_1 + \theta_2) + L_1 \cos(\theta_1) & L_2 \cos(\theta_1 + \theta_2) & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad (8)$$

c)

$$\mathbf{J} = \begin{bmatrix} -L_2 \sin(\theta_1 + \theta_2) - L_1 \sin(\theta_1) & L_2 \cos(\theta_1 + \theta_2) + L_1 \cos(\theta_1) & 0 \\ -L_2 \sin(\theta_1 + \theta_2) & L_2 \cos(\theta_1 + \theta_2) & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad (9)$$

Question-6

What would be the joint torques needed to exert a force vector $\mathbf{f} = [1 \ 1 \ 1]^T$ at the tip of the 3-link manipulator with the homogeneous transformation matrix:

$${}^0\mathbf{T}_4 = \begin{bmatrix} \cos(\theta_1 + \theta_2 + \theta_3) & -\sin(\theta_1 + \theta_2 + \theta_3) & 0 & L_1 \cos(\theta_1) + L_2 \cos(\theta_1 + \theta_2) \\ \sin(\theta_1 + \theta_2 + \theta_3) & \cos(\theta_1 + \theta_2 + \theta_3) & 0 & L_1 \sin(\theta_1) + L_2 \sin(\theta_1 + \theta_2) \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (10)$$

a)

$$\boldsymbol{\tau} = \begin{bmatrix} L_2 \sin(\theta_1 + \theta_2) + L_1 \sin(\theta_1) + L_2 \cos(\theta_1 + \theta_2) + L_1 \cos(\theta_1) \\ -L_2 \sin(\theta_1 + \theta_2) + L_2 \cos(\theta_1 + \theta_2) \\ 0 \end{bmatrix} \quad (11)$$

b)

$$\boldsymbol{\tau} = \begin{bmatrix} -L_2 \sin(\theta_1 + \theta_2) - L_1 \sin(\theta_1) + L_2 \cos(\theta_1 + \theta_2) + L_1 \cos(\theta_1) \\ -L_2 \sin(\theta_1 + \theta_2) + L_2 \cos(\theta_1 + \theta_2) \\ 0 \end{bmatrix} \quad (12)$$

c)

$$\boldsymbol{\tau} = \begin{bmatrix} -L_2 \sin(\theta_1 + \theta_2) - L_1 \sin(\theta_1) - L_2 \sin(\theta_1 + \theta_2) \\ L_2 \cos(\theta_1 + \theta_2) + L_1 \cos(\theta_1) + L_2 \cos(\theta_1 + \theta_2) \\ 0 \end{bmatrix} \quad (13)$$

Question-7

What condition in joint angles in the 3-link manipulator with the homogeneous transformation matrix given by

$${}^0T_4 = \begin{bmatrix} \cos(\theta_1 + \theta_2 + \theta_3) & -\sin(\theta_1 + \theta_2 + \theta_3) & 0 & L_1 \cos(\theta_1) + L_2 \cos(\theta_1 + \theta_2) \\ \sin(\theta_1 + \theta_2 + \theta_3) & \cos(\theta_1 + \theta_2 + \theta_3) & 0 & L_1 \sin(\theta_1) + L_2 \sin(\theta_1 + \theta_2) \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (14)$$

will result in a torque only around joint-1 to exert a force vector $\mathbf{f} = [1 \ 1 \ 1]^T$ at the tip?

- a) $\theta_2 = 0$
- b) $\theta_2 + \theta_3 = \pi/2$
- c) $\theta_2 + \theta_3 = \pi/4$
- d) $\theta_1 + \theta_2 = \pi/4$

Question-8

Which of the following vectors is in a homogeneous coordinate notation?

a)

$$\mathbf{v}_1 = [2 \ 4 \ 1 \ 0]^T \quad (15)$$

b)

$$\mathbf{v}_2 = [4 \ 8 \ 2 \ 2]^T \quad (16)$$

c)

$$\mathbf{v}_3 = [4i \ 8j \ 2k]^T \quad (17)$$

Question-9

- a) What is the purpose of the first rotation of a $Z - Y - X$ Euler rotation?
- b) Likewise, what is the purpose of the second rotation?

Question-10

How many sequences are there for Euler rotations? i.e. $Z - Y - X$, $X - Y - Z$, ...

- a) For moving frame rotations
- b) For fixed frame rotations

$\times \neq$

$$\begin{aligned} \cos\left(\frac{\theta}{2} + \frac{\theta}{2}\right) &= \cos\frac{\theta}{2} \cdot \cos\frac{\theta}{2} - \sin\frac{\theta}{2} \sin\frac{\theta}{2} \\ &= \cos^2\frac{\theta}{2} - \sin^2\frac{\theta}{2} \end{aligned}$$